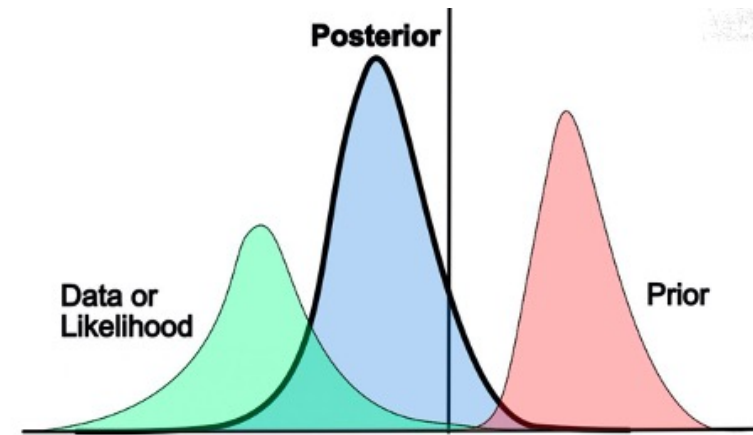




INFERENCES

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Inferences

using logic and reasoning to draw conclusions or make predictions about unknown information based on available data, evidence, or observations.



Inferences

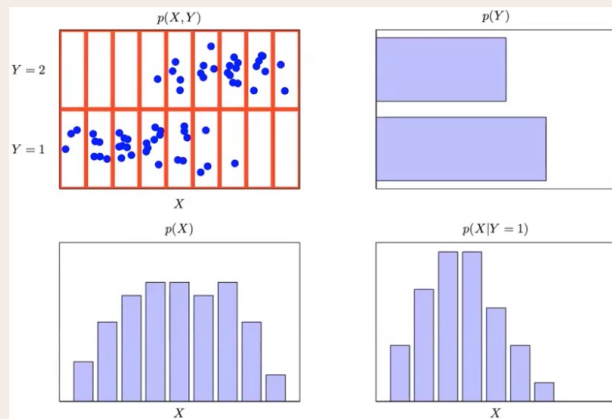
using logic and reasoning to draw conclusions or make predictions about unknown information based on available data, evidence, or observations.

Deduction	Induction	Abduction
A conclusion that logically follows from a set of premises, with absolute certainty.	A conclusion based on specific observations, with some degree of uncertainty.	A conclusion based on the best explanation for a set of observations, with some degree of uncertainty.
All men are mortal. Socrates is a man. Therefore, Socrates is mortal. – Aristotle	A turkey found that every day for the past year he was fed at 9am sharp. Therefore, he concludes that “I am always fed at 9am”. However, on the morning of Christmas eve, he had his throat cut. – Bertrand Russell	I had grasped the significance of the silence of the dog, for one true inference invariably suggests others.... Obviously, the midnight visitor was someone whom the dog knew well. – <The Adventure of Silver Blaze>
If a star's mass is above 10 solar masses, it will end its life in a supernova explosion. R136a1 has a mass of 200 solar masses. Therefore, R136a1 will end its life in a supernova explosion.	Many observed exoplanets have been found to orbit close to their host stars. It's likely that the newly discovered exoplanet Kepler-452b also orbits close to its star.	We observe the circular velocity of stars and gas in many galaxies remains flat in the outskirts of galaxies. Dark matter!



Bayesian Inference

updating our understanding
based on new evidence



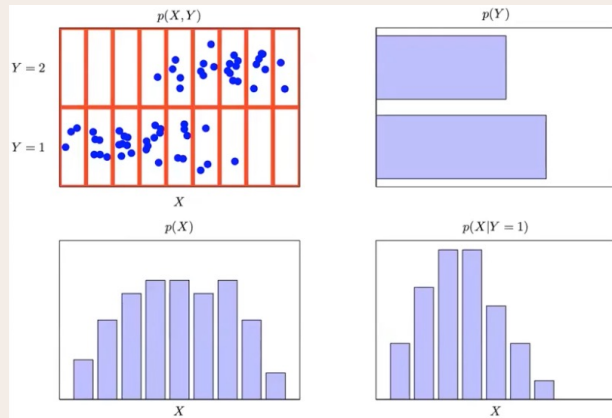
- Joint probability $P(X, Y)$: the probability of X and Y
- Marginal probability $P(X)$ or $P(Y)$: the probability of X or Y
- Conditional probability $P(X|Y)$ or $P(Y|X)$: the probability of X given Y or of Y given X
- Sum rule, $P(X) = \sum_Y P(X, Y)$, $P(Y) = \sum_X P(X, Y)$
- Product rule, $P(X, Y) = P(X|Y)P(Y) = P(Y|X)P(X)$: the probability of X and Y equals to the probability of X (or Y) times the probability of Y given X (or of X given Y)
- Bayes' rule: $P(X|Y) = P(Y|X)P(X)/P(Y)$

Recap



Bayesian Inference

updating our understanding
based on new evidence



- Joint probability $P(X, Y)$: the probability of X and Y
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Recap

X: a manufactured component is defective

Y: the component fails a quality-control test

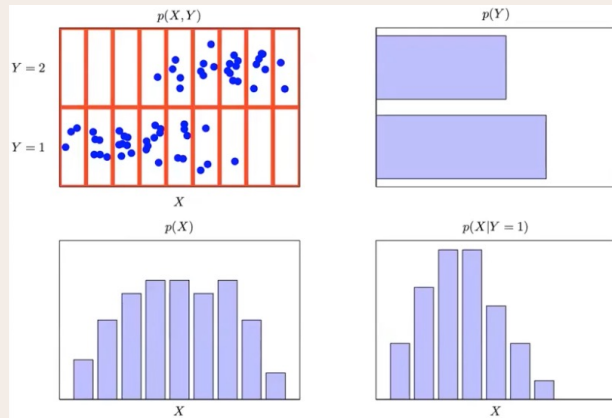
Say 2% of components produced by a factory are defective, 95% of defective components fail the quality-control test, and 10% of all components fail the test.

given that a component failed the test, there is a probability that it is actually defective.



Bayesian Inference

updating our understanding
based on new evidence



- Joint probability $P(X, Y)$: the probability of X and Y
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- Bayes' rule: $P(X|Y) = P(Y|X)P(X)/P(Y)$

Recap

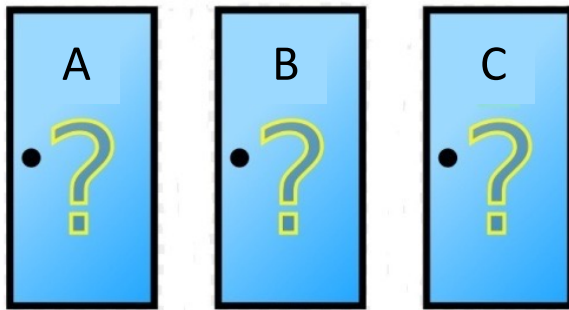
X: a manufactured component is defective
Y: the component fails a quality-control test
 Say 2% of components produced by a factory are defective, 95% of defective components fail the quality-control test, and 10% of all components fail the test.

$$P(\text{defective} \mid \text{fails test}) = \frac{P(\text{defective})P(\text{fails test} \mid \text{defective})}{P(\text{fails test})}$$

given that a component failed the test, there is a 19% probability that it is actually defective.



The Monty Hall problem



$$P(A|C) = P(C|A)P(A)/P(C)$$

$P(A)$: the probability of the prize behind Door A (1/3)

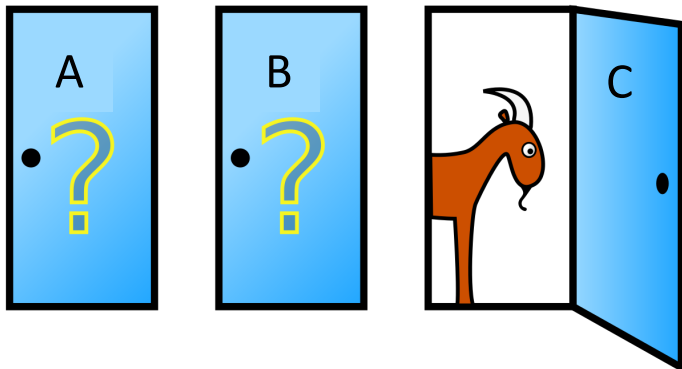
-> Prior

$P(C|A)$: If the prize is behind Door A, the probability of the host opening Door C is 0.5

$P(C)$: the probability of the host opening Door C:

- $P(C | A) = 0.5$
- $P(C | \text{not } A)$:
 - $P(C | \text{If the prize is behind door B}) = 1$
 - $P(C | \text{If the prize is behind door C}) = 0.$

$$P(C) = (0.5 + 1 + 0) / 3 = 0.5$$



$$P(A|C) = P(C|A)P(A)/P(C) = 1/3 \quad \text{-> Posterior}$$



Bayesian Inference

updating our understanding
based on new evidence

X: your model parameters θ
Y: the observed data D

Say you look at the data points $(x_1, x_2, \dots, x_N)$
and find that they seem to follow a normal
distribution: $\theta \equiv (\mu, \sigma)$

$$P(D|\theta) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} e^{-\left(\frac{x_i - \mu}{\sqrt{2}\sigma}\right)^2}$$

Bayes' rule: $P(\theta|D) = P(D|\theta)P(\theta)/P(D)$

- Joint probability $P(X, Y)$: the probability of X and Y
- Marginal probability $P(X)$ or $P(Y)$: the probability of X or Y
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- Bayes' rule: $P(X|Y) = P(Y|X)P(X)/P(Y)$

Recap

- $P(D|\theta)$: the likelihood function, representing the probability of observing the data (D) given a specific set of modeling parameters (θ)
- $P(\theta)$: the prior probability distribution over the modeling parameters (θ). This represents our initial beliefs of possible values of θ
- $P(D) \equiv \sum_{\theta} P(D, \theta) = \sum_{\theta} P(\theta)P(D|\theta) = \int d\theta P(\theta)P(D|\theta)$: the marginal likelihood, representing the overall probability of observing the data (D)
- $P(\theta|D)$: the posterior probability distribution over the modeling parameters (θ) given the data.



Bayesianism vs Frequentism

To a frequentist, probability is not a degree of belief. It describes the behaviour of a random process under hypothetical repeated sampling.

To a Bayesian, probability can also represent uncertainty or degree of belief about an unknown quantity or event.

Example 1: When a frequentist is asked to determine the probability of flipping a coin to land on the head, they will keep flipping the coin many many times, collect data and calculate the frequency of head in the end of the experiment. But a Bayesian statistician will first guess e.g., 50% based on their prior knowledge of coin flipping and keep update the probability based on what they observe when flipping the coin

Example 2: Suppose I have not flipped a fair coin yet and ask, “What is the probability that it will land heads?” Both a frequentist and a Bayesian can say 50%. For the frequentist, 50% describes the long-run frequency of heads over repeated flips. For the Bayesian, 50% also represents their uncertainty about the outcome of the upcoming flip.

Example 3: Now suppose I flip the fair coin and hide the result. A Bayesian can still say there is a 50% probability that the hidden result is heads, because probability can represent uncertainty about an unknown outcome. For a strict frequentist, the outcome is already fixed—either heads or tails—so probability no longer represents uncertainty about that particular realised outcome.



Bayesianism vs Frequentism

To a frequentist, probability is not a degree of belief. It describes the behaviour of a random process under hypothetical repeated sampling.

To a Bayesian, probability can also represent uncertainty or degree of belief about an unknown quantity or event.

Example 4: Higgs boson discovery at the LHC

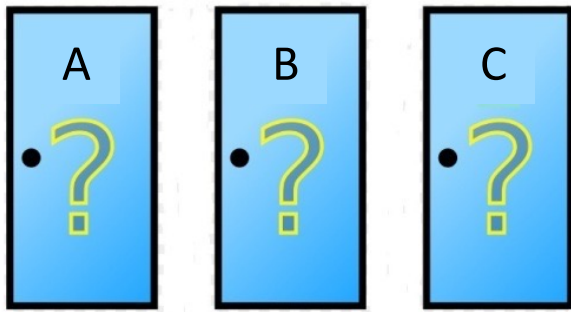
Physicists test a null hypothesis such as “there is no Higgs boson signal, only background.” They ask: if the null hypothesis were true, how often would repeated experiments produce data at least this extreme?

This gives a p-value, often reported as a significance in units of σ . A 5σ result corresponds to a one-sided p-value of about 2.9×10^{-7} , meaning that background alone would produce an excess this large only about 3 times in 10 million repeated experiments.

The key frequentist idea is: $P(\text{"data this extreme or more"} | H_0)$ not $P(H_0 | \text{data})$. So when CERN reported a 5σ Higgs discovery, it did not mean: “There is a 99.99997% probability that the Higgs exists.” Instead, it meant roughly: “If there were no Higgs signal, data this signal-like would be extraordinarily rare under repeated experiments.”



Monty Hall meets frequentists



```
def host_open_door(my_choice, win_door, door_options):  
    # Determine which door the host opens  
    for door in door_options:  
        # cannot open my choice or the winning door  
        if door != my_choice and door != win_door:  
            return door  
  
def my_switches(host_door, my_choice, door_options):  
    # Simulate the contestant switching doors  
    for door in door_options:  
        # cannot be the original choice or the host door  
        if door != my_choice and door != host_door:  
            return door
```

```
door_options = ["A", "B", "C"]  
win_by_switching = []  
win_by_staying = []  
  
# Simulate many games of Monty Hall  
for game_num in range(10000):  
    # Randomly choose the winning door  
    win_door = random_choice(door_options)  
  
    # Randomly choose a door for the contestant to initially choose  
    my_choice = random_choice(door_options)  
  
    # Host opens one of the other two doors that is not the winning one  
    host_door = host_open_door(my_choice, win_door, door_options)  
  
    # contestant switches doors if they want  
    switch_door = switches(host_door, my_choice, door_options)  
  
    if switch_door == win_door:  
        win_by_switching.append(1) # Switching won this game  
        win_by_staying.append(0)  # Staying lost this game  
    else:  
        win_by_switching.append(0) # Switching lost this game  
        win_by_staying.append(1)  # Staying won this game  
  
# Calculate the probabilities of winning  
prob_win_by_switching = sum(win_by_switching)/10000  
prob_win_by_staying   = sum(win_by_staying)/10000
```



Bayesian inference pseudocode

Say you look at the data points $(x_1, x_2, \dots x_N)$ and find that they seem to follow a normal distribution: $\theta \equiv (\mu = 5, \sigma)$

```
# Initial prior and samples
prior = initial_prior
std_samples = sample_parameters(initial_prior, std_min, std_max, num_samples)
mean_guessed = 5

prior_samples      = np.zeros(num_samples)
likelihood_samples = np.zeros(num_samples)
posterior_samples  = np.zeros(num_samples)

# Inference loop
for istep in range(num_steps):
    for i in range(num_samples):
        prior_samples[i] = prior(std_samples[i])
        likelihood_samples[i] = np.prod(norm_pdf(data, mean_guessed, std_samples[i]))
        posterior_samples[i] = likelihood_samples[i] * prior_samples[i]

    marginal_likelihoods = np.sum(likelihood_samples) / num_samples
    posterior_samples /= marginal_likelihoods

# Interpolate posterior for next step's prior
f_interp = interp1d(std_samples, posterior_samples)
def prior(mean, std):
    return f_interp(mean, std)

std_samples = sample_parameters(prior, std_min, std_max, num_samples)
```

$$P(\theta|D) = P(D|\theta)P(\theta)/P(D)$$

$P(D|\theta)$: likelihood
 $P(\theta)$: prior probability
 $P(D)$: marginal likelihood
 $P(\theta|D)$: posterior probability

Points in a sample might only have a uniform volume when initialized to be uniform and are very likely not the case at later times!



prerequisite

git clone/pull from <https://github.com/qyx268/astr4004-8004-2026.git>

Install scipy, emcee, dynesty, corner

Install torch, sbi, swyft



open inference-part1.ipynb



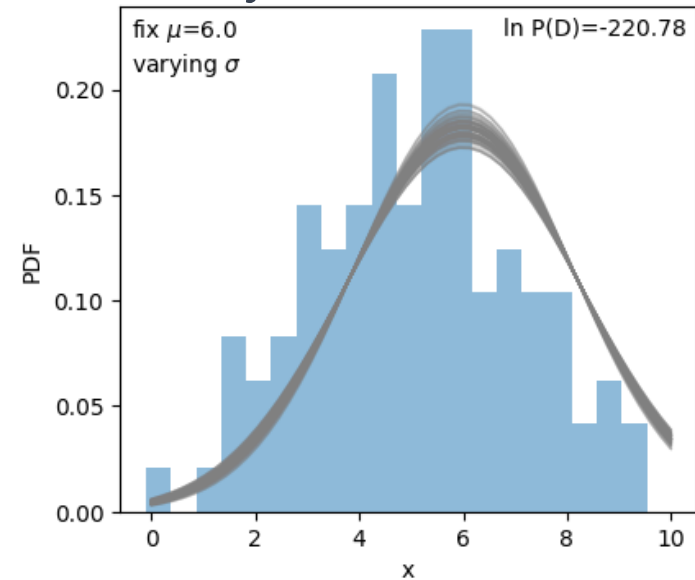
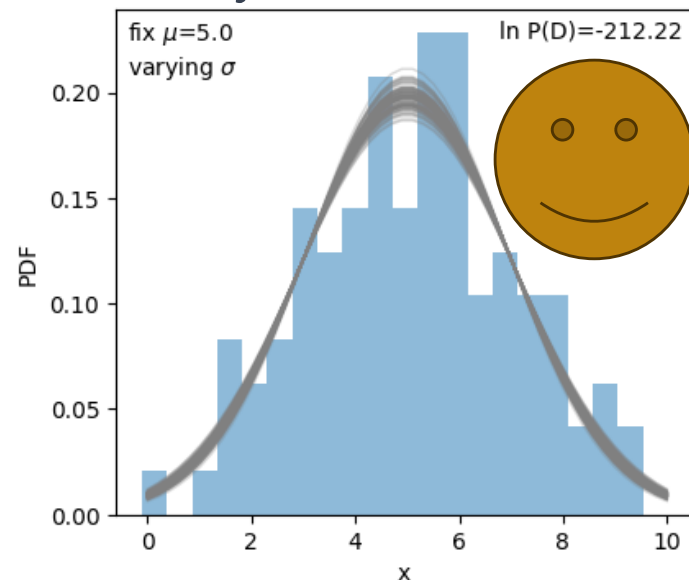
Bayesian evidence

$$P(D) \equiv \sum_{\theta} P(D, \theta) = \sum_{\theta} P(\theta)P(D|\theta) = \int d\theta P(\theta)P(D|\theta):$$

The marginal likelihood, representing the overall probability of observing the data (D)

$$P(D|\mathcal{M}_1) = \int d\theta P(\theta|\mathcal{M}_1)P(D|\theta, \mathcal{M}_1)$$

$$P(D|\mathcal{M}_2) = \int d\theta P(\theta|\mathcal{M}_2)P(D|\theta, \mathcal{M}_2)$$



A higher evidence indicates the model is more likely to have generated the observed data.

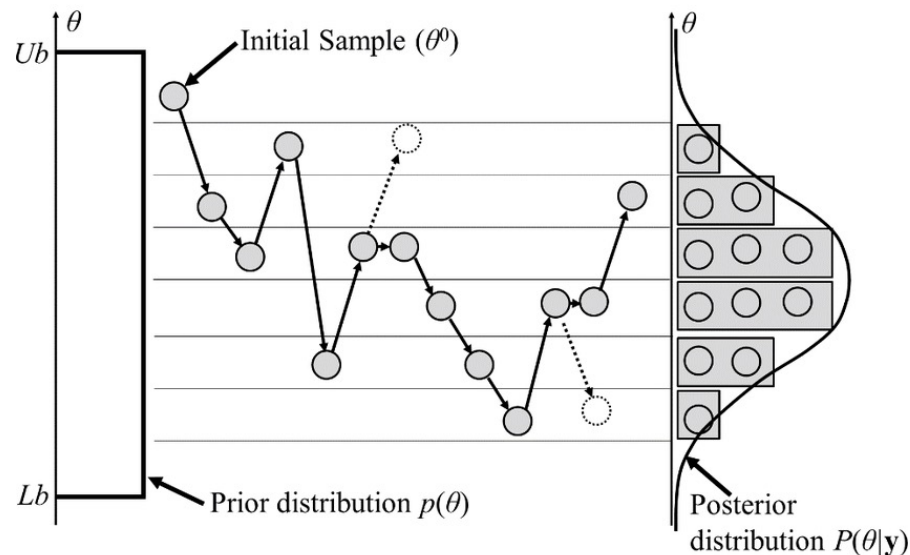


open inference-part2.ipynb



Metropolis-Hastings Algorithm

- Generate a set of samples θ
- Draw a new sample θ_{new} according to some local proposal distribution $Q(\theta_{\text{new}}|\theta_{\text{old}})$
- Compute the likelihood ratio $r = P(\theta_{\text{new}})/P(\theta_{\text{old}})$
- If $r \geq 1$, accept the proposal; otherwise, accept it with a probability of r



When reaching equilibrium

$$P(\theta_{\text{old}}) Q(\theta_{\text{new}}|\theta_{\text{old}}) A(\theta_{\text{new}}|\theta_{\text{old}}) = P(\theta_{\text{new}}) Q(\theta_{\text{old}}|\theta_{\text{new}}) A(\theta_{\text{old}}|\theta_{\text{new}})$$

Q is symmetric as $Q(\theta_{\text{new}}|\theta_{\text{old}}) = Q(\theta_{\text{old}}|\theta_{\text{new}})$

When the acceptance probability is defined as $A = \min(r, 1)$, the equilibrium is naturally enforced and $A \in [0, 1]$.

$$r \equiv \frac{P(\theta_{\text{new}})}{P(\theta_{\text{old}})} = \frac{A(\theta_{\text{new}}|\theta_{\text{old}})}{A(\theta_{\text{old}}|\theta_{\text{new}})}$$

$$= \frac{\min(r, 1)}{\min(1/r, 1)} = \begin{cases} r & \text{if } r < 1 \\ 1/r & \text{otherwise} \end{cases} = r$$

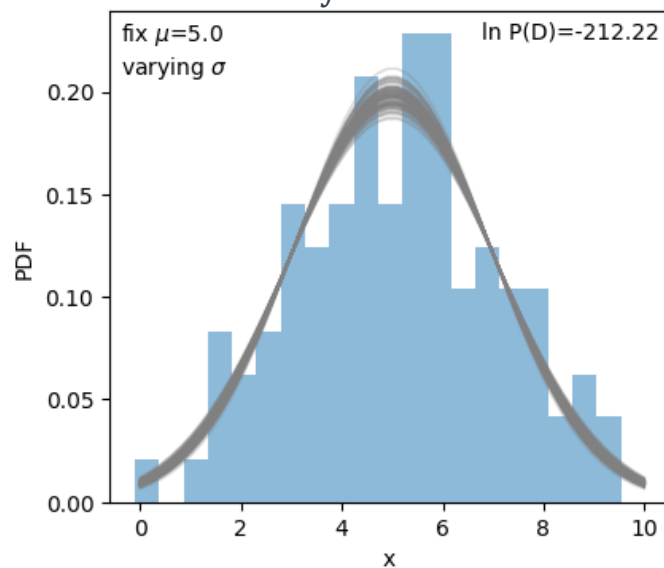


Bayesian evidence

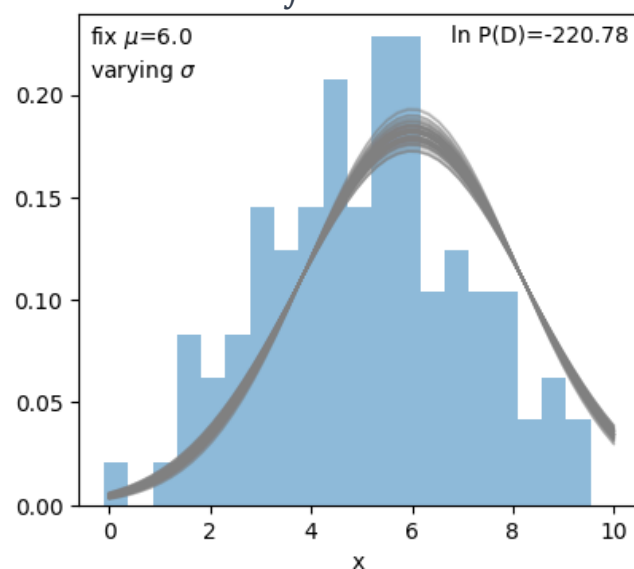
$$P(D) \equiv \sum_{\theta} P(D, \theta) = \sum_{\theta} P(\theta)P(D|\theta) = \int d\theta P(\theta)P(D|\theta):$$

The marginal likelihood, representing the overall probability of observing the data (D)

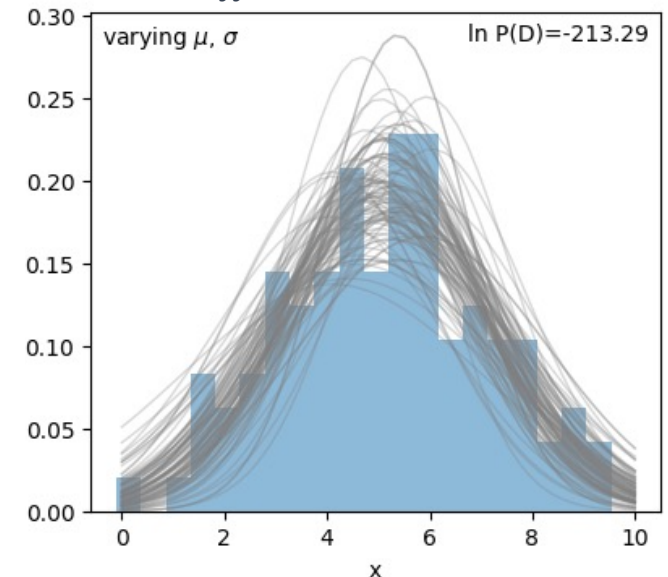
$$P(D) = \int d\sigma P(\sigma)P(D|\sigma)$$



$$P(D) = \int d\sigma P(\sigma)P(D|\sigma)$$



$$P(D) = \iint d\mu d\sigma P(\mu, \sigma)P(D|\mu, \sigma)$$



A higher evidence indicates the model is more likely to have generated the observed data.



Bayesian evidence

Occam's razor: introducing additional parameters penalizes the evidence unless preferred by the data

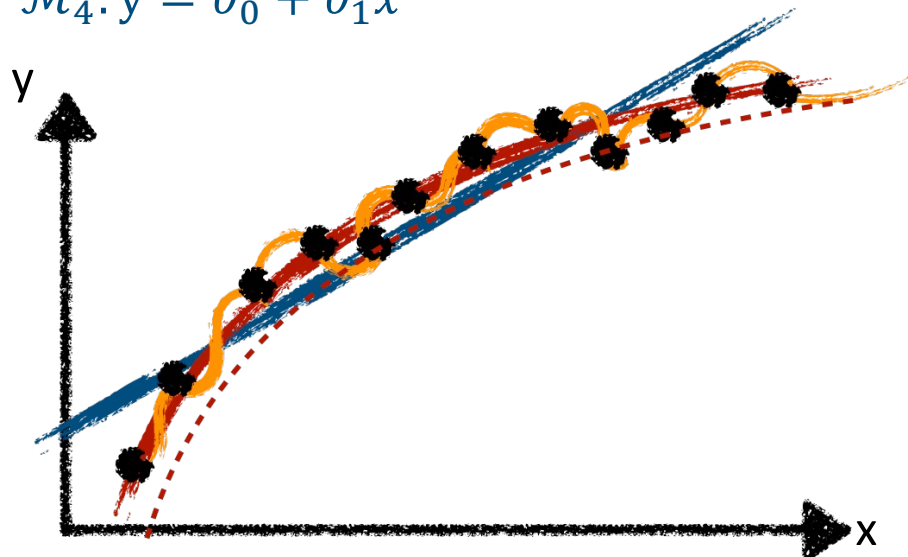
$P(D) = \int d\theta P(\theta)P(D|\theta)$: the marginal likelihood, representing the overall probability of observing the data (D)

$$\mathcal{M}_1: y = -1 + \theta_1 x + \theta_2 x^2$$

$$\mathcal{M}_2: y = -2 + \theta_1 x + \theta_2 x^2$$

$$\mathcal{M}_3: y = \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_n x^n$$

$$\mathcal{M}_4: y = \theta_0 + \theta_1 x$$

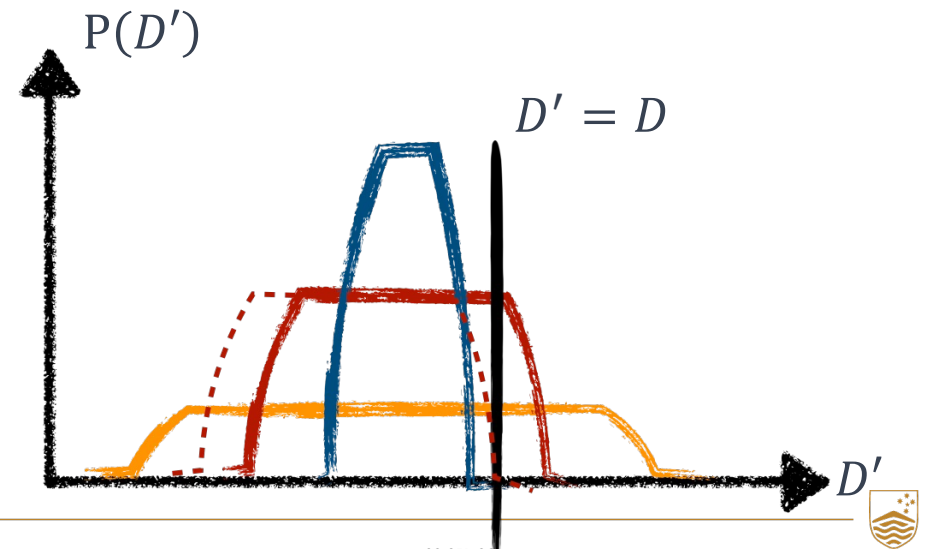


A good model with the right number of free parameters

A poor model with the same number of free parameters

A very complex model that is able to predict different possible dataset but with low probability

A model that is too simple to explain the observation



open inference-part3.ipynb



Nested sampling

$$P(\theta|D) = P(D|\theta)P(\theta)/P(D)$$

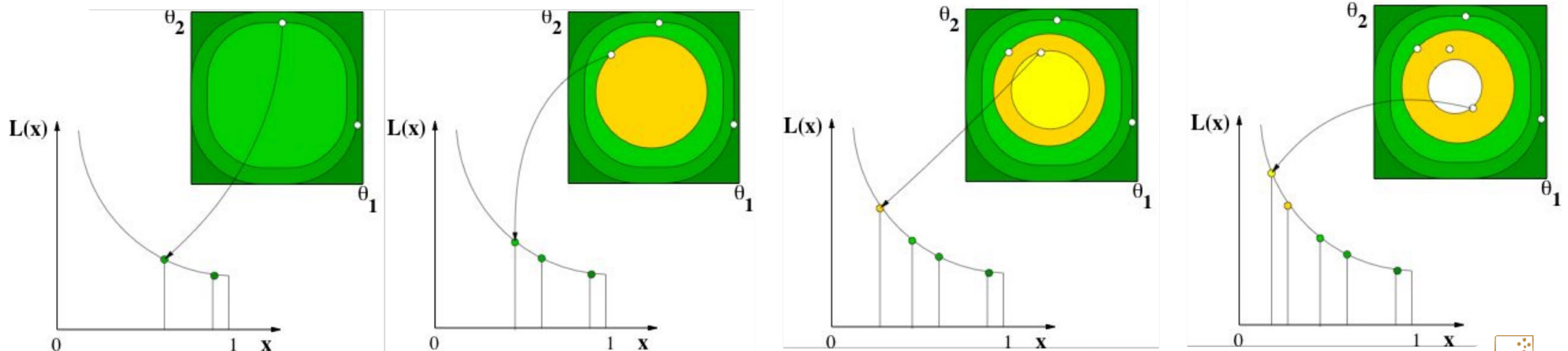
$P(D|\theta)$: likelihood
 $P(\theta)$: prior probability
 $P(D)$: marginal likelihood
 $P(\theta|D)$: posterior probability

- We define iso-likelihood contour, which encapsulates/nests parameter space (θ) where likelihoods, $L(\theta) \equiv P(D|\theta)$, are higher than λ .
- We re-write our prior volume as:

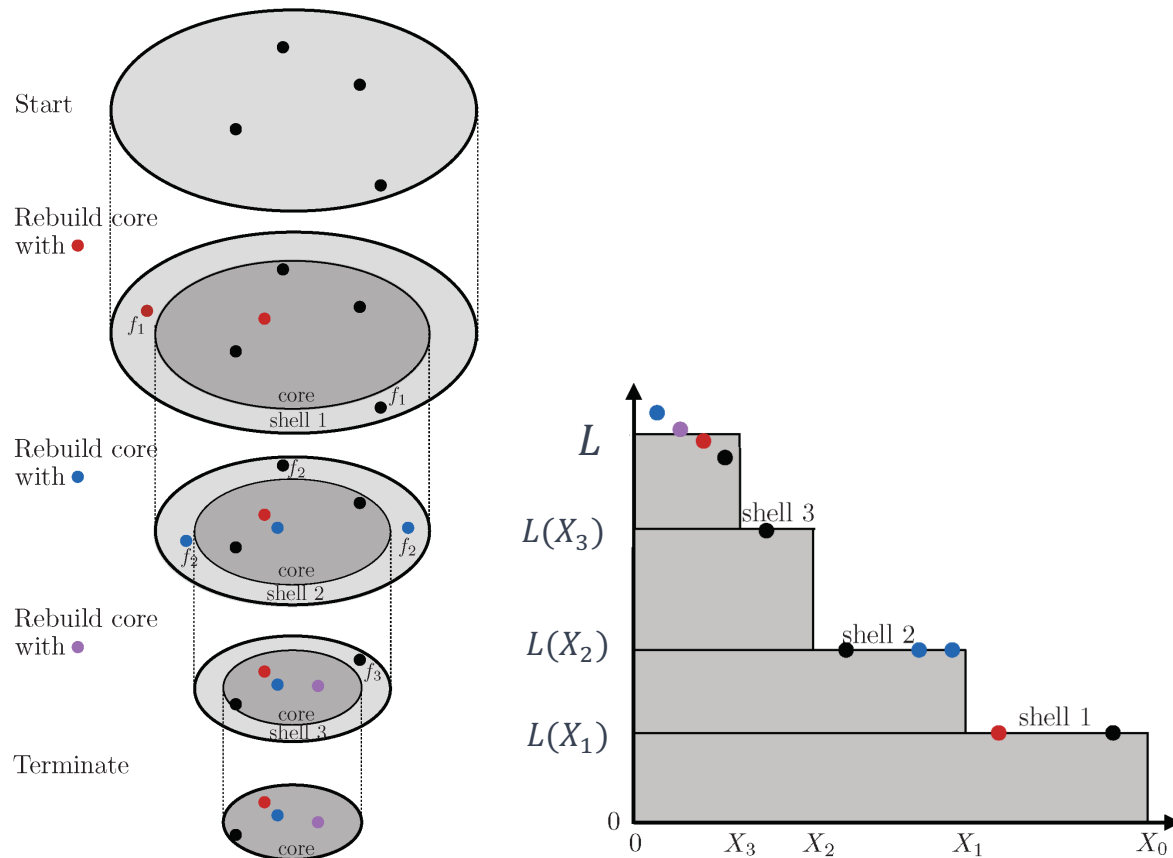
$$X(\lambda) = \int_{L(\theta) > \lambda} d\theta P(\theta)$$

- We re-write the evidence in the space of prior volume, $X(\lambda)$:

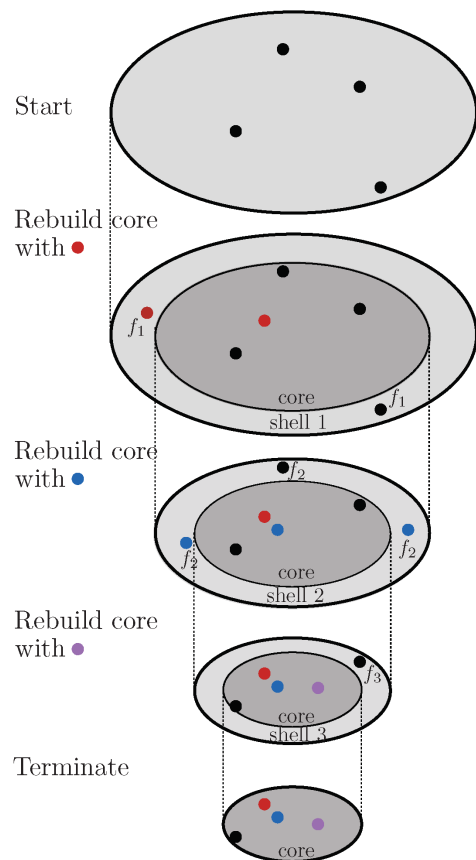
$$Z = P(D) = \int_{\theta} d\theta P(D|\theta)P(\theta) = \int_0^1 dX L(X) = \sum L(X_i) \Delta X_i$$



Nested sampling pseudocode



Nested sampling pseudocode



open inference-part4.ipynb



Likelihood-free inference

Simulation-based inference (SBI)

$$P(\theta|D) = P(D|\theta)P(\theta)/P(D)$$

$P(D|\theta)$: likelihood

$P(\theta)$: prior probability

$P(D)$: marginal likelihood

$P(\theta|D)$: posterior probability

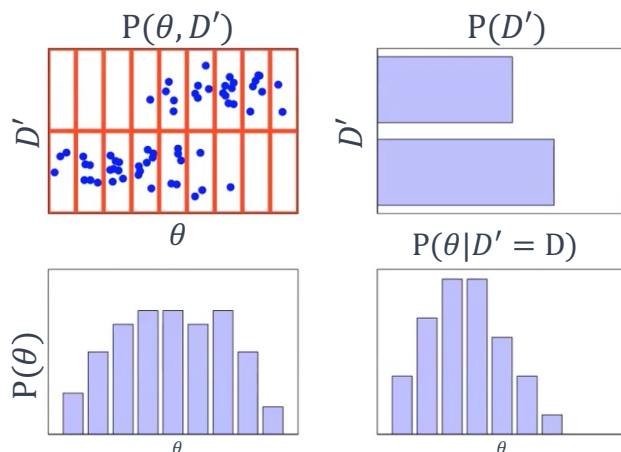
What if we don't know the likelihood or there is no functional form for the likelihood?

Remember our model can reproduce different outputs (D') for given θ . This means that we can measure the frequency of those outputs that reproduce the real data (D).

Approximate Bayesian Computation

```
def approximate_bayesian_computation(observed_data, num_samples, tolerance):  
    accepted_parameters = []
```

```
    for i in range(num_samples):  
        # 1. Sample a parameter theta from the prior  
        theta = sample_prior()  
  
        # 2. Simulate data using the parameter theta  
        simulated_data = simulate_data(theta)  
  
        # 3. Compare simulated data to observed data  
        distance = calculate_distance(simulated_data, observed_data)  
  
        # 4. Accept or reject the parameter based on the distance  
        if distance < tolerance: accepted_parameters.append(theta)  
  
    return accepted_parameters
```



Likelihood-free inference

Simulation-based inference (SBI)

$$P(\theta|D) = P(D|\theta)P(\theta)/P(D)$$

$P(D|\theta)$: likelihood

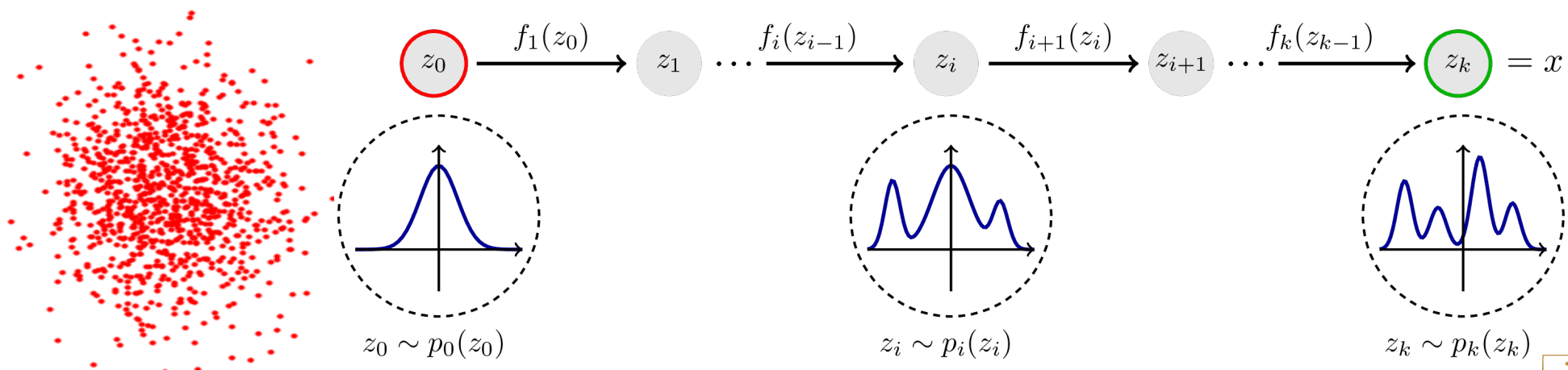
$P(\theta)$: prior probability

$P(D)$: marginal likelihood

$P(\theta|D)$: posterior probability

What if we can use a few sample of (θ, D') to estimate $P(\theta|D)$?

Normalizing flow: A chain of invertible transformations constituting a normalizing flow, converting a simple distribution $p_0(z_0)$ into a complex one $p_k(z_k)$ step by step. The flows are trained to fit $p_k(z_k)$ to the target distribution $p(x)$, e.g., $P(\theta|D)$, $P(D|\theta)$, $P(D, \theta)$, ...



open inference-part5.ipynb



Likelihood-free inference

Simulation-based inference (SBI)

$$P(\theta|D) = P(D|\theta)P(\theta)/P(D)$$

$P(D|\theta)$: likelihood

$P(\theta)$: prior probability

$P(D)$: marginal likelihood

$P(\theta|D)$: posterior probability

What if we can use a few sample of (θ, D') to estimate the ratio of $P(\theta|D)$ to $P(\theta)$?

Marginal Neural Ratio Estimation

Recall the product rule, $P(\theta, D) = P(\theta|D)P(D) = P(D|\theta)P(\theta)$

$$r(\theta, D) \equiv \frac{P(\theta, D)}{P(\theta)P(D)} = \frac{P(\theta|D)}{P(\theta)} = \frac{P(D|\theta)}{P(D)}$$

Introduce a binary random variable y corresponding to whether (θ, D) is drawn jointly ($y = 1$) or marginally/independently ($y = 0$) and this joint distribution is $\tilde{P}(\theta, D, y)$

$$r(\theta, D) \equiv \frac{\tilde{P}(\theta, D|y=1)}{\tilde{P}(\theta, D|y=0)} = \frac{\frac{\tilde{P}(\theta, D, y=1)}{\tilde{P}(y=1)}}{\frac{\tilde{P}(\theta, D, y=0)}{\tilde{P}(y=0)}} = \frac{\tilde{P}(\theta, D, y=1)}{\tilde{P}(\theta, D, y=0)} = \frac{\frac{\tilde{P}(\theta, D, y=1)}{\tilde{P}(\theta, D)}}{\frac{\tilde{P}(\theta, D, y=0)}{\tilde{P}(\theta, D)}} = \frac{\tilde{P}(y=1|\theta, D)}{\tilde{P}(y=0|\theta, D)} = \frac{\tilde{P}(y=1|\theta, D)}{1 - \tilde{P}(y=1|\theta, D)}$$

Now the goal becomes training a classifier $\tilde{P}(y=1|\theta, D)$, which informs the likelihood-to-evidence ratio or posterior-to-prior ratio.



open inference-part6.ipynb

